Arthur K. MacKeith
Curriculum Vitae

Contact information and digital portfolio at `Arthur.Mackeith.info`

EDUCATION

Ph.D., Mechanical Engineering & Materials Science, Yale University Expected May 2027

Advisor: Professor Corey S. O'Hern
Research topics: Morphogenesis of leaf spongy mesophyll tissue in three dimensions, efficient methods for deformable particles in two and three dimensions

M.S. & M.Phil. en route to Ph.D. May 2024

Selected Course Work: Mathematical Methods of Physics, Statistical Physics I & II, Biological Physics, Biomedical Image Processing & Analysis

M.S., Computer Science, The University of Chicago June 2020

B.A., Physics Joint degree with M.S.

Cumulative GPA: 3.69/4.00,

Selected Course Work: Computer Science: Math Foundations of Machine Learning, Introduction to Machine Learning, Algorithms, Computer Architecture, Parallel Programming, Time Series Analysis and Stochastic Processes. Physics: Honors Mechanics, Intermediate Mechanics, Mathematical Methods of Physics, Geophysical Fluid Dynamics I and II, Statistical Models and Methods, Statistical and Thermal Physics.

Honors and Awards:
James Franck Institute Summer Research Fellowship (2018)
Dean's List (2016-2019)

PUBLICATIONS

5. **Arthur K. MacKeith**, Dong Wang, Mark D. Shattuck, and Corey S. O'Hern. "Deformable Particles: Modeling and Applications." In *Packing Problems in Soft Matter Physics*, edited by Ho-Kei Chan, Stefan Hutzler, Adil Mughal, Corey S. O'Hern, Yujie Wang, and Denis Weaire, vol. 27. Theoretical and Computational Chemistry Series. Royal Society of Chemistry, 2025.

4. Sven Witthaus, Atoosa Parsa, Dong Wang, Nidhi Pashine, Jerry Zhang, **Arthur K. MacKeith**, Mark D. Shattuck, Josh Bongard, Corey S. O'Hern, and Rebecca Kramer-Bottiglio. "Evolution of adaptive force chains in reconfigurable granular metamaterials." *Soft Matter* 21, no. 30 (2025): 6088-6099.

3. Andrew T. Ton, **Arthur K. MacKeith**, Mark D. Shattuck, and Corey S. O'Hern. "Mechanical plasticity of cell membranes enhances epithelial wound closure." *Physical Review Research* 6, no. 1 (2024): L012036.

2. Won-Kyu Lee, Daniel J. Preston, Markus P. Nemitz, Amit Nagarkar, **Arthur K. MacKeith**, Benjamin Gorissen, Nikolas Vasios, Vanessa Sanchez, Katia Bertoldi, L. Mahadevan, and George M. Whitesides. "A buckling-sheet ring oscillator for electronics-free, multimodal locomotion." *Science Robotics* 7, no. 63 (2022): eabg5812.

1. Kieran A. Murphy, **Arthur K. MacKeith**, Leah K. Roth, and Heinrich M. Jaeger. "The intertwined roles of particle shape and surface roughness in controlling the shear strength of a granular material." *Granular Matter* 21, no. 3 (2019): 72.

PRESENTATIONS

3. **Invited:** Arthur K. MacKeith, Adam B. Roddy, Mark D. Shattuck, and Corey S. O'Hern. "Modeling Mesophyll Tissue Morphogenesis in Three Dimensions." *Gordon Research Seminar: Interdisciplinary Approaches to Soft and Living Matter*. 2025.
2. **Contributed:** Arthur K. MacKeith, Adam B. Roddy, Mark D. Shattuck, Corey S. O'Hern. "Modeling Spongy Mesophyll Development." *American Physical Society: Joint March Meeting and April Meeting*. 2025.
1. **Contributed:** Arthur K. MacKeith, Allison E. Culbert, John D. Treado, Adam B. Roddy, Craig Brodersen, Mark D. Shattuck, and Corey S. O'Hern. "Modeling and Analysis of Mesophyll Tissue Development in Leaves and Flowers Across Species." *American Physical Society: March Meeting*. 2023.

RESEARCH EXPERIENCE PRIOR TO PH.D.

Robotic Intelligence through Perception Lab, Toyota Technological Institute at Chicago, Chicago IL

Research Engineer August 2019 to June 2021

- Explored multidimensional design and control spaces for soft robots with Sofa Framework simulation (finite element method (FEM) software)
- Built interface between reinforcement learning algorithms and Sofa Framework
- Optimized FEM soft robot model for simulation speed and machine learning
- Integrated a low-cost drone into education, and research robotics platform Duckietown. These drones are currently being used in public high schools across Rhode Island for hands on STEM education

George M. Whitesides Group, Harvard University, Cambridge MA

REU Student June 2019 to August 2019

- Designed and built a low-cost semi-soft beam climbing robot, incorporating soft controls such that it only requires a single constant input pressure gas source to climb
- Exploited static asymmetric friction to produce climbing from contracting actuators and optimized the asymmetry such that the robot was able to climb with a payload equal to its own weight (70 g)
- Designed a pneumatic signal processing demonstration and characterized it using Matlab

Jaeger Lab, The University of Chicago, Chicago IL

Research Assistant December 2017 to June 2019

- Wrote and optimized image analysis python program that fit the position and orientation of the several thousand particles in a granular sample with 99.5% success rate

– Analyzed position and orientation data in python for global trends and mesoscale structures
– Designed 3D particles to print that break predictably to detect shifts in packing structure
– Recorded and reconstructed computed tomography scans of packings of granular media

OUTREACH

Girls' Science Investigations, Yale Pathways, 2022-Present

Physics with a Bang! Holiday Lecture and Open House, 2018 - Volunteer Tour Guide

Duckietown Robotics Teaching Platform (See Research Experience) - Developer

SKILLS

– Proficient Languages: Python, C++, Matlab, CUDA
– Languages with some experience: C, Go, DrRacket (LISP), MIPS Assembly Language
– Tools and Operating Systems: ROS, Docker, Sofa Framework (Finite element method software), Solidworks (some experience), Swig (C and C++ wrapper for Python and other languages), Ubuntu, OSX
– Hardware: 3D printing, Silicon molding, Soldering, Woodworking, Instron